12.25 The Largest of Many

WORKSHEET

The question

STEP III 2025, Question 11

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The largest of $X_1, \ldots, X_n$ is at most y exactly when what happens?

STAGE 1 The distribution of the maximum

Covers: Part (i)

Ask yourself The largest of the X_i is at most y exactly when every X_i is at most y. Use independence.

STAGE 2 The quantiles

Covers: Part (ii)

Ask yourself Solve $\mathrm{G}(L) = \alpha$ for L, and $\mathrm{G}(U) = 1 - \alpha$ for U.

STAGE 3 Large n

Covers: Part (iii)

Pause and try Write $\alpha^{1/n} = e^{(\ln \alpha)/n}$. For large n the exponent is small, so use $e^t \approx 1 + t$.

STAGE 4 Growth and spread

Covers: Part (iv)

Ask yourself The median is $L(\frac{1}{2})$: what happens to it as $n \to \infty$? Subtract your approximations for U and L. Which terms cancel?

STAGE 5 **A 90% interval**

Covers: Part (v)

Ask yourself Take $\alpha = 0.05$. Approximate $\ln 20$ and $\ln \frac{1}{0.95}$.

What's really going on

The Gumbel distribution

Since $\mathrm{P}(\lambda Y - \ln n \le x) = (1 - \frac{e^{-x}}{n})^n \to \exp(-e^{-x})$, the shifted maximum $\lambda Y - \ln n$ has, in the limit, the *Gumbel distribution* with distribution function $\exp(-e^{-x})$. Its mean is Euler's constant $\gamma \approx 0.577$. You met it already: the time to collect all n coupons (Section 9.11) is the maximum of n (nearly independent) waiting times, which is why it is about $n \ln n + \gamma n$, with Gumbel fluctuations of order n.

Extreme value theory

A remarkable theorem of Fisher, Tippett and Gnedenko (1928–1943) says that, suitably shifted and scaled, the maximum of many independent identically distributed variables can have only three possible limiting shapes: Gumbel (for light tails, like the exponential or normal), Fréchet (for heavy tails, like the Cauchy) and Weibull (for variables with a hard upper limit). It is the extreme-value analogue of the central limit theorem (Section 12.24). Engineers use it to design flood defences and offshore platforms for the "hundred-year wave", insurers to price catastrophe cover, and climate scientists to estimate how heatwaves are changing.

Gumbel and the floods

Emil Julius Gumbel, a German statistician and an outspoken opponent of the Nazis, who was forced out of his university post in 1932 and eventually fled to the United States, applied extreme value theory to the flooding of rivers in the 1930s and 1940s. His book *Statistics of Extremes* (1958) made the subject a practical tool. The Dutch Delta Works, built after the North Sea flood of 1953 to protect the Netherlands, were designed using extreme value statistics to withstand a storm surge expected about once in 10 000 years.

Try this next

1. **The expected maximum.** Show that $\mathrm{E}(Y) = \frac{1}{\lambda}(1 + \frac{1}{2} + \cdots + \frac{1}{n})$. (*Hint:* by memorylessness, after the first of the n lifetimes ends, the remaining $n-1$ start afresh.)

2. **The minimum.** Show that the smallest of $X_1, \ldots, X_n$ is exponential with parameter $n\lambda$.

3. **The median exactly.** Find the exact median of Y and compare it with the approximation $\frac{1}{\lambda}(\ln n - \ln \ln 2)$ for $n = 10$.

4. **Uniform maxima.** If $X_1, \ldots, X_n$ are uniform on $[0,1]$, find the distribution of their maximum M, and show that $n(1-M)$ is approximately exponential for large n. Which of the three extreme-value shapes is this?

Further reading

- **Stuart Coles, *An Introduction to Statistical Modeling of Extreme Values*** (Springer). The standard introduction, with real data on floods and sea levels.
- **Emil Julius Gumbel, *Statistics of Extremes*** (Dover). The classic.
- **David Spiegelhalter, *The Art of Uncertainty*** (Pelican), on rare events and how to think about them.

The full guided solution is in Section 12.25:
https://beyondstep.keithmcnulty.org/ch12/extremes.html